

Integration — Lesson 17 Test: Substitution I (Linear Insides)

10 questions · show your work in the box · check every antiderivative by differentiating · don't forget + C

Name: _____ Date: _____ Score: _____ / 10

1. Find $\int (x+3)^4 dx$.

1 pt

2. Find $\int (2x+5)^2 dx$, and check your answer by differentiating it.

1 pt

3. Find $\int e^{(3x)} dx$.

1 pt

4. Find $\int \cos(4x) dx$.

1 pt

5. Find $\int \sin(3x) dx$. (Watch both the minus sign and the divide.)

1 pt

6. A student writes $\int e^{(5x)} dx = e^{(5x)} + C$. Differentiate their answer to show the mistake, then write the correct answer.

1 pt

7. Compute $\int_0^1 (6x+2)^2 dx$. (Tip: evaluate the inside $6x+2$ at each limit first.)

1 pt

8. Compute $\int_0^\pi \cos(x/2) dx$. (Remember: dividing by $1/2$ means multiplying by 2.)

1 pt

9. Compute $\int_0^\pi \sin(2x) \, dx$, and explain the answer using signed area.

1 pt

10. Challenge: compute $\int_0^1 (e^{4x} + (3x+2)^2) \, dx$. Give the exact answer.

1 pt
